CSE 318 — Online 4

Section B1

You have generated neighbiors by reversing one segment of the tour, disconnecting and reconnecting two edges. Update that portion so that it cuts the tour at three positions and reconnects the three segments in a new way. There are several reconnection variants; you will implement one specific and simple variant below.

What to implement.
Write `generateNeighbor3Opt(tour)` that returns one neighbor tour by the following 3-cut move. Let N be the number of cities. City 0 is at position 0 and stays there.

1. Pick three positions $i < j < k$ uniformly at random with $1 \leq i < j < k \leq N - 1$.
2. Split the tour into four parts:
   - $A =$ `tour[0 ...i-1]`
   - $B =$ `tour[i ...j-1]`
   - $C =$ `tour[j ...k-1]`
   - $D =$ `tour[k ...N-1]`
3. Return the concatenation

$$A \ + \ \text{reverse}(B) \ + \ \text{reverse}(C) \ + \ D.$$

   (Both middle segments are reversed.)

Plug this in place of your current neighbor generator (or use it as an additional option — but pick one and be consistent). Keep everything else (initial solution, temperature, cooling, acceptance) identical to your original SA.

Output:

- Run modified SA **5 times** on **at least two** instances. Report **best, average, worst** cost.
- Report the total number of accepted moves and worse-accepted moves.